

9-2c Units-of-Activity Method

The **units-of-activity method** (A method of depreciation that provides the same amount of depreciation expense for each unit of an asset's activity, which may be expressed in hours, miles driven, or quantity produced.)

provides the same amount of depreciation expense for each unit of activity of the asset. Depending on the asset, the units of activity can be expressed in terms of hours, miles driven, or quantity produced. For example, the unit of activity for a truck is normally expressed in miles driven. For manufacturing assets, the units of activity are often expressed as units of product. In this case, the units-of-activity method may be called the *units-of-production method* or *units-of-output method*.

The units-of-activity method is applied in the following two steps:

Step 1.

Determine the depreciation per unit as follows:

$$\text{Depreciation per Unit} = \frac{\text{Cost} - \text{Residual Value}}{\text{Total Estimated Units of Activity}}$$

Step 2.

Compute the depreciation expense as follows:

$$\text{Depreciation Expense} = \text{Depreciation per Unit} \times \text{Units of Activity for Period}$$

To illustrate, assume that **Exeter's** forklift is estimated to have a useful life of 10,000 operating hours over the life of the asset. During the first year, the forklift was operated 2,100 hours. The units-of-activity depreciation for the year is \$4,620, computed as follows:

Step 1.

Determine the depreciation per hour as follows:

$$\text{Depreciation per Hour} = \frac{\text{Cost} - \text{Residual Value}}{\text{Total Estimated Units of Activity}} = \frac{\$24,000 - \$2,000}{10,000 \text{ Hours}} = \$2.20 \text{ per Hour}$$

Step 2.

Compute the depreciation expense as follows:

$$\begin{aligned} \text{Depreciation Expense} &= \text{Depreciation per Unit} \times \text{Units of Activity for Period} \\ &= \$2.20 \text{ per Hour} \times 2,100 \text{ Hours} = \$4,620 \end{aligned}$$

Depreciation for the first year using the units-of-activity method is recorded as follows:

Dec. 31	Depreciation Expense—Forklift		4,620	
	Accumulated Depreciation—Forklift			4,620

Assets	=	Liabilities	+	Stockholders' Equity
Accumulated Depreciation				Depreciation Expense
4,620				4,620

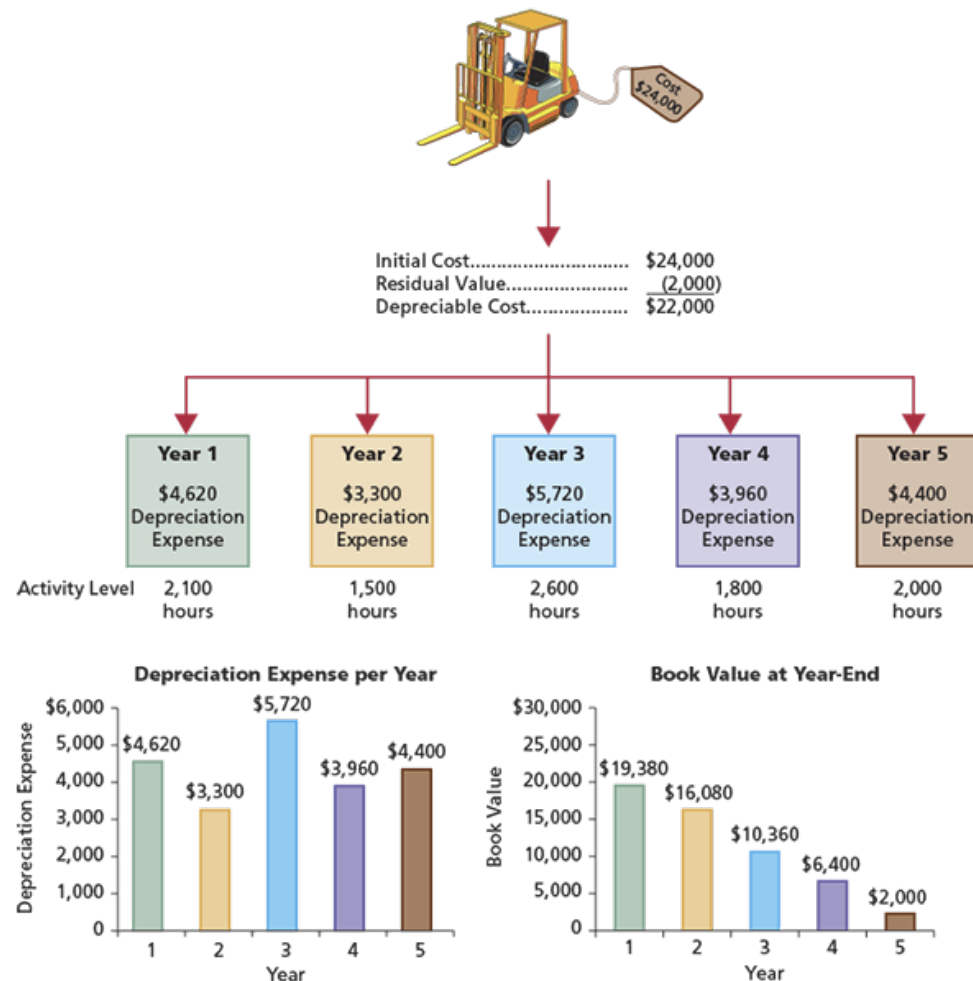
Assume that during its five-year life, the forklift was used as follows:

Year 1	2,100 hours
Year 2	1,500
Year 3	2,600
Year 4	1,800
Year 5	<u>2,000</u>
Total	<u>10,000 hours</u>

Exhibit 6 illustrates the depreciation expense and book value of the forklift over its five-year life using the units-of-activity method.

Exhibit 6

Units-of-Activity Method



As shown in Exhibit 6, depreciation expense and book value vary each year depending on the hours the forklift is operated. The fixed asset's book value will equal its residual value at the end of the asset's useful life.

The units-of-activity method is often used when a fixed asset's use varies from year to year. In such cases, the units-of-activity method matches depreciation expense with the asset's revenues.